Take Home Activity 05

Q1.

HotelRoom.java > HotelRoom > main(String[])

```java
class HotelRoom {
    int roomNumber;
    String roomType;
    double pricePerNight;
    boolean isBooked;

    HotelRoom() {
        roomNumber = 0;
        roomType = "Standard";
        pricePerNight = 5000.0;
        isBooked = false;
    }

    HotelRoom(int roomNumber, String roomType, double pricePerNight, boolean isBooked) {
        this.roomNumber = roomNumber;
        this.roomType = roomType;
        this.pricePerNight = pricePerNight;
        this.isBooked = isBooked;
    }

    double bookRoom(int nights) {
        if (!isBooked) {
            isBooked = true;
            return pricePerNight * nights;
        } else {
            System.out.println("Room is already booked.");
            return 0.0;
        }
    }

    void checkoutRoom() {
        isBooked = false;
        System.out.println("Room " + roomNumber + " is now available.");
    }

    String getRoomSummary() {
        return "Room " + roomNumber + " - " + roomType +
               " - Rs. " + pricePerNight +
               " per night - Booked: " + isBooked;
    }

```

```java
    Run | Debug
    public static void main(String[] args) {
        HotelRoom room1 = new HotelRoom();

        HotelRoom room2 = new HotelRoom(roomNumber: 101, roomType: "Deluxe", pricePerNight: 8500.0, isBooked: false);

        double totalCost = room2.bookRoom(nights: 3);
        System.out.println("Total Cost for 3 nights: Rs. " + totalCost);

        room2.bookRoom(nights: 2);

        room2.checkoutRoom();

        double newBooking = room2.bookRoom(nights: 2);
        System.out.println("New Booking Cost for 2 nights: Rs. " + newBooking);

        System.out.println(x: "\nRoom Summaries:");
        System.out.println(room1.getRoomSummary());
        System.out.println(room2.getRoomSummary());
    }
}
```

```
D:\NSBM\NSBM\Year 1\Semester 2\Java\Assessment\Assessment 5>java H
Total Cost for 3 nights: Rs. 25500.0
Room is already booked.
Room 101 is now available.
New Booking Cost for 2 nights: Rs. 17000.0

Room Summaries:
Room 0 - Standard - Rs. 5000.0 per night - Booked: false
Room 101 - Deluxe - Rs. 8500.0 per night - Booked: true
```

Q2.

OnlineOrder.java > OnlineOrder > main(String[])

```java
class OnlineOrder {
    int orderId;
    String customerName;
    double itemPrice;
    int quantity;

    OnlineOrder(int orderId, String customerName, double itemPrice, int quantity) {
        this.orderId = orderId;
        this.customerName = customerName;
        this.itemPrice = itemPrice;
        this.quantity = quantity;
    }

    double calculateTotal(double discountPercentage) {
        double total = itemPrice * quantity;
        double discount = total * (discountPercentage / 100);
        return total - discount;
    }

    void updateQuantity(int newQuantity) {
        if (newQuantity > 0) {
            quantity = newQuantity;
            System.out.println("Quantity updated to " + newQuantity + ".");
        } else {
            System.out.println(x: "Invalid quantity.");
        }
    }

    String getOrderSummary() {
        return "Order ID: " + orderId +
               ", Customer Name: " + customerName +
               ", Item Price: Rs. " + itemPrice +
               ", Quantity: " + quantity;
    }

    Run | Debug
    public static void main(String[] args) {
        OnlineOrder order1 = new OnlineOrder(orderId: 101, customerName: "Kasun Perera", itemPrice: 2500.0, quantity: 2);
        OnlineOrder order2 = new OnlineOrder(orderId: 102, customerName: "Nimal Silva", itemPrice: 1500.0, quantity: 5);

        double finalAmount = order1.calculateTotal(discountPercentage: 10);
        System.out.println("Final Amount after 10% discount: Rs. " + finalAmount);

        order1.updateQuantity(newQuantity: 4);

        order2.updateQuantity(-2);

        System.out.println(x: "\nOrder Summaries:");
        System.out.println(order1.getOrderSummary());
        System.out.println(order2.getOrderSummary());
    }
}
```

```
D:\NSBM\NSBM\Year 1\Semester 2\Java\Assessment\Assessment 5>java OnlineOrder.java
Final Amount after 10% discount: Rs. 4500.0
Quantity updated to 4.
Invalid quantity.

Order Summaries:
Order ID: 101, Customer Name: Kasun Perera, Item Price: Rs. 2500.0, Quantity: 4
Order ID: 102, Customer Name: Nimal Silva, Item Price: Rs. 1500.0, Quantity: 5
```

Q3.

StudentResult.java > StudentResult > StudentResult(int, String, double, double, double)

```java
class StudentResult {
    int studentId;
    String studentName;
    double mathMark;
    double scienceMark;
    double englishMark;

    StudentResult(int studentId, String studentName, double mathMark, double scienceMark, double englishMark) {

        this.studentId = studentId;
        this.studentName = studentName;
        this.mathMark = mathMark;
        this.scienceMark = scienceMark;
        this.englishMark = englishMark;
    }

    double calculateAverage() {
        return (mathMark + scienceMark + englishMark) / 3;
    }

    String getGrade() {
        double average = calculateAverage();

        if (average >= 75) {
            return "A";
        } else if (average >= 65) {
            return "B";
        } else if (average >= 50) {
            return "C";
        } else if (average >= 35) {
            return "S";
        } else {
            return "F";
        }
    }

    void updateMark(String subject, double newMark) {

        if (subject.equalsIgnoreCase(anotherString: "math")) {
            mathMark = newMark;
        } else if (subject.equalsIgnoreCase(anotherString: "science")) {
            scienceMark = newMark;
        } else if (subject.equalsIgnoreCase(anotherString: "english")) {
            englishMark = newMark;
        } else {
            System.out.println(x: "Invalid subject name.");
        }
    }

    String getResultSheet() {
        return "Student ID: " + studentId +
                "\nStudent Name: " + studentName +
                "\nMath Mark: " + mathMark +
                "\nScience Mark: " + scienceMark +
                "\nEnglish Mark: " + englishMark +
                "\nAverage: " + calculateAverage() +
                "\nGrade: " + getGrade();
    }

```

```java
    }

    Run | Debug
    public static void main(String[] args) {
        StudentResult student1 = new StudentResult(studentId: 101, studentName: "Kamal", mathMark: 80, scienceMark: 75, … 70);

        StudentResult student2 = new StudentResult(studentId: 102, studentName: "Nimali", mathMark: 55, scienceMark: 60, … 45);

        student1.updateMark(subject: "math", newMark: 90);
        student2.updateMark(subject: "english", newMark: 65);

        student1.updateMark(subject: "history", newMark: 50);

        System.out.println(x: "----- Student 1 Result Sheet -----");
        System.out.println(student1.getResultSheet());

        System.out.println(x: "\n----- Student 2 Result Sheet -----");
        System.out.println(student2.getResultSheet());
    }
}
```

```
D:\NSBM\NSBM\Year 1\Semester 2\Java\Assessment\Asses
Invalid subject name.
----- Student 1 Result Sheet -----
Student ID: 101
Student Name: Kamal
Math Mark: 90.0
Science Mark: 75.0
English Mark: 70.0
Average: 78.33333333333333
Grade: A

----- Student 2 Result Sheet -----
Student ID: 102
Student Name: Nimali
Math Mark: 55.0
Science Mark: 60.0
English Mark: 65.0
Average: 60.0
Grade: C
```

Q4.

VehicalRental.java > VehicleRental > main(String[])

```java
class VehicleRental {
    int vehicleId;
    String vehicleType;
    double ratePerDay;
    double availableKm;
    boolean isAvailable;

    VehicleRental() {
        vehicleId = 0;
        vehicleType = "Car";
        ratePerDay = 5000.0;
        availableKm = 300.0;
        isAvailable = true;
    }

    VehicleRental(int vehicleId, String vehicleType,
                  double ratePerDay, double availableKm,
                  boolean isAvailable) {

        this.vehicleId = vehicleId;
        this.vehicleType = vehicleType;
        this.ratePerDay = ratePerDay;
        this.availableKm = availableKm;
        this.isAvailable = isAvailable;
    }

    String rentVehicle(int days, double estimatedKm) {
        if (!isAvailable) {
            return "Vehicle is currently unavailable.";
        }

        if (estimatedKm > availableKm) {
            return "Requested kilometers exceed available kilometers.";
        }

        isAvailable = false;
        double totalCost = ratePerDay * days;

        return "Vehicle rented successfully.\n" + "Total Cost: Rs. " + totalCost;
    }

    void returnVehicle(double kmUsed) {
        availableKm -= kmUsed;
        isAvailable = true;

        System.out.println(x: "Vehicle returned successfully.");
        System.out.println("Remaining Available KM: " + availableKm);
        System.out.println(x: "Vehicle is now available.");
    }

    String getVehicleInfo() {

        return "Vehicle ID: " + vehicleId +
               "\nVehicle Type: " + vehicleType +
               "\nRate Per Day: Rs. " + ratePerDay +
               "\nAvailable KM: " + availableKm +
               "\nAvailable Status: " + isAvailable;
    }

```

```java
class VehicleRental {

    Run | Debug
    public static void main(String[] args) {
        VehicleRental vehicle1 = new VehicleRental();

        VehicleRental vehicle2 = new VehicleRental(vehicleId: 101, vehicleType: "Van", ratePerDay: 8000.0, availableK… 500.0, true);

        System.out.println(x: "----- Vehicle 1 Info -----");
        System.out.println(vehicle1.getVehicleInfo());

        System.out.println(x: "\n----- Vehicle 2 Info -----");
        System.out.println(vehicle2.getVehicleInfo());

        System.out.println(x: "\n--- Renting Vehicle 2 ---");
        System.out.println(vehicle2.rentVehicle(days: 3, estimatedKm: 200));

        System.out.println(x: "\nVehicle Info After Renting:");
        System.out.println(vehicle2.getVehicleInfo());

        System.out.println(x: "\n--- Trying to Rent Again ---");
        System.out.println(vehicle2.rentVehicle(days: 2, estimatedKm: 100));

        System.out.println(x: "\n--- Returning Vehicle ---");
        vehicle2.returnVehicle(kmUsed: 150);

        System.out.println(x: "\nVehicle Info After Return:");
        System.out.println(vehicle2.getVehicleInfo());

        System.out.println(x: "\n--- Renting Again ---");
        System.out.println(vehicle2.rentVehicle(days: 2, estimatedKm: 100));

        System.out.println(x: "\nFinal Vehicle Info:");
        System.out.println(vehicle2.getVehicleInfo());
    }
}
```

```
D:\NSBM\NSBM\Year 1\Semester 2\Java\Assessment\Assessment 5>java V
----- Vehicle 1 Info -----
Vehicle ID: 0
Vehicle Type: Car
Rate Per Day: Rs. 5000.0
Available KM: 300.0
Available Status: true

----- Vehicle 2 Info -----
Vehicle ID: 101
Vehicle Type: Van
Rate Per Day: Rs. 8000.0
Available KM: 500.0
Available Status: true

--- Renting Vehicle 2 ---
Vehicle rented successfully.
Total Cost: Rs. 24000.0

Vehicle Info After Renting:
Vehicle ID: 101
Vehicle Type: Van
Rate Per Day: Rs. 8000.0
Available KM: 500.0
Available Status: false

--- Trying to Rent Again ---
Vehicle is currently unavailable.

--- Returning Vehicle ---
Vehicle returned successfully.
Remaining Available KM: 350.0
Vehicle is now available.

Vehicle Info After Return:
Vehicle ID: 101
Vehicle Type: Van
Rate Per Day: Rs. 8000.0
Available KM: 350.0
Available Status: true

--- Renting Again ---
Vehicle rented successfully.
Total Cost: Rs. 16000.0

Final Vehicle Info:
Vehicle ID: 101
Vehicle Type: Van
Rate Per Day: Rs. 8000.0
Available KM: 350.0
Available Status: false
```

Q5.

LibraryBook.java > LibraryBook > main(String[])

```java
class LibraryBook {
    int bookId;
    String title;
    String author;
    int totalCopies;
    int borrowedCopies;

    LibraryBook() {
        bookId = 0;
        title = "Unknown Title";
        author = "Unknown Author";
        totalCopies = 5;
        borrowedCopies = 0;
    }

    LibraryBook(int bookId, String title, String author, int totalCopies) {
        this.bookId = bookId;
        this.title = title;
        this.author = author;
        this.totalCopies = totalCopies;
        this.borrowedCopies = 0;
    }

    String borrowBook(int count) {
        int available = totalCopies - borrowedCopies;

        if (count <= available) {
            borrowedCopies += count;
            available = totalCopies - borrowedCopies;

            return "Successfully borrowed " + count +
                    " copy/copies. Remaining: " + available;
        } else {
            return "Only " + available + " copy/copies available.";
        }
    }

    void returnBook(int count) {
        if (count <= borrowedCopies) {
            borrowedCopies -= count;

            System.out.println(count +
                     " copy/copies returned successfully.");
        } else {
            System.out.println(x: "Return count exceeds borrowed copies.");
        }
    }

    int getAvailableCopies() {
        return totalCopies - borrowedCopies;
    }
```

```java

    String getBookInfo(String shelf) {
        return "Book ID: " + bookId +
                "\nTitle: " + title +
                "\nAuthor: " + author +
                "\nTotal Copies: " + totalCopies +
                "\nBorrowed Copies: " + borrowedCopies +
                "\nAvailable Copies: " + getAvailableCopies() +
                "\nShelf Location: " + shelf;
    }

    Run | Debug
    public static void main(String[] args) {
        LibraryBook book1 = new LibraryBook();

        LibraryBook book2 = new LibraryBook(bookId: 101, title: "Java Programming", author: "James Gosling", totalCopies: 10);

        System.out.println(x: "----- Book 1 Info -----");
        System.out.println(book1.getBookInfo(shelf: "A1"));

        System.out.println(x: "\n----- Book 2 Info -----");
        System.out.println(book2.getBookInfo(shelf: "B2"));

        System.out.println(x: "\n--- Borrowing Books ---");
        System.out.println(book2.borrowBook(count: 4));

        System.out.println(x: "\nBook Info After Borrowing:");
        System.out.println(book2.getBookInfo(shelf: "B2"));

        System.out.println(x: "\n--- Borrowing More Than Available ---");
        System.out.println(book2.borrowBook(count: 7));

        System.out.println(x: "\n--- Returning Books ---");
        book2.returnBook(count: 2);

        System.out.println(x: "\nBook Info After Returning:");
        System.out.println(book2.getBookInfo(shelf: "B2"));

        System.out.println(x: "\n--- Borrowing Again ---");
        System.out.println(book2.borrowBook(count: 3));

        System.out.println(x: "\nFinal Book Info:");
        System.out.println(book2.getBookInfo(shelf: "B2"));
    }
}
```

```
D:\NSBM\NSBM\Year 1\Semester 2\Java\Assessment\Assessment 5>java LibraryBook.
----- Book 1 Info -----
Book ID: 0
Title: Unknown Title
Author: Unknown Author
Total Copies: 5
Borrowed Copies: 0
Available Copies: 5
Shelf Location: A1

----- Book 2 Info -----
Book ID: 101
Title: Java Programming
Author: James Gosling
Total Copies: 10
Borrowed Copies: 0
Available Copies: 10
Shelf Location: B2

--- Borrowing Books ---
Successfully borrowed 4 copy/copies. Remaining: 6

Book Info After Borrowing:
Book ID: 101
Title: Java Programming
Author: James Gosling
Total Copies: 10
Borrowed Copies: 4
Available Copies: 6
Shelf Location: B2

--- Borrowing More Than Available ---
Only 6 copy/copies available.

--- Returning Books ---
2 copy/copies returned successfully.

Book Info After Returning:
Book ID: 101
Title: Java Programming
Author: James Gosling
Total Copies: 10
Borrowed Copies: 2
Available Copies: 8
Shelf Location: B2

--- Borrowing Again ---
Successfully borrowed 3 copy/copies. Remaining: 5

Final Book Info:
Book ID: 101
Title: Java Programming
Author: James Gosling
Total Copies: 10
Borrowed Copies: 5
Available Copies: 5
Shelf Location: B2
```